

Our residuals are this:

$$S = \sum_{p \neq q} \sum_{mn} d_{pq}^{mn} - g_p^{ma} M^{ab} \overline{g_q^{bn}}. \quad (1)$$

p and q are baselines, m and n are correlations ($\in r, l$). Repeated indices are summed. What we minimize is the norm of S , that is, $S\overline{S}$, or $|S|^2$.

As “selfcalibration”, we use the solves Jones matrices and the data used to solve the Jones matrices, to solve for model. We do this to see if there is an model differences. It will also indicate the performance of our fitting solution. We do this by evaluating the gradient and using Gradient descent. The same way we solved for Jones matrices.

So, we compute the derivative below: All summations $\sum$ over all the baselines.

$$\begin{aligned} \frac{\partial |S|^2}{\partial \overline{m^{rr}}} = & -d^{lr} g_q^{rr} \overline{g_p^{lr}} + \\ & -d^{ll} g_q^{rl} \overline{g_p^{lr}} + \\ & -d^{rr} g_q^{rr} \overline{g_p^{rr}} + \\ & -d^{rl} g_q^{rl} \overline{g_p^{rr}} + \\ & m^{rl} \left(g_p^{lr} g_q^{rl} \overline{g_p^{lr} g_q^{ll}} + g_p^{lr} g_q^{rr} \overline{g_p^{lr} g_q^{lr}} + g_p^{rr} g_q^{rl} \overline{g_p^{rr} g_q^{ll}} + g_p^{rr} g_q^{rr} \overline{g_p^{rr} g_q^{lr}} \right) + \\ & m^{ll} \left(g_p^{ll} g_q^{rl} \overline{g_p^{ll} g_q^{ll}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{lr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rr} g_q^{ll}} + g_p^{rl} g_q^{rr} \overline{g_p^{rr} g_q^{lr}} \right) + \\ & m^{rr} \left(g_p^{lr} g_q^{rl} \overline{g_p^{lr} g_q^{rl}} + g_p^{lr} g_q^{rr} \overline{g_p^{lr} g_q^{rr}} + g_p^{rr} g_q^{rl} \overline{g_p^{rr} g_q^{rl}} + g_p^{rr} g_q^{rr} \overline{g_p^{rr} g_q^{rr}} \right) + \\ & m^{lr} \left(g_p^{ll} g_q^{rl} \overline{g_p^{ll} g_q^{rl}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rr} g_q^{rl}} + g_p^{rl} g_q^{rr} \overline{g_p^{rr} g_q^{rr}} \right). \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial m^{rl}} = & -d^{lr} g_q^{lr} \overline{g_p^{lr}} + \\ & -d^{ll} g_q^{ll} \overline{g_p^{lr}} + \\ & -d^{rr} g_q^{lr} \overline{g_p^{rr}} + \\ & -d^{rl} g_q^{ll} \overline{g_p^{rr}} + \\ & m^{rl} \left(g_p^{lr} g_q^{ll} \overline{g_p^{lr} g_q^{ll}} + g_p^{lr} g_q^{lr} \overline{g_p^{lr} g_q^{lr}} + g_p^{rr} g_q^{ll} \overline{g_p^{rr} g_q^{ll}} + g_p^{rr} g_q^{lr} \overline{g_p^{rr} g_q^{lr}} \right) + \\ & m^{ll} \left(g_p^{ll} g_q^{ll} \overline{g_p^{ll} g_q^{ll}} + g_p^{ll} g_q^{lr} \overline{g_p^{ll} g_q^{lr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rr} g_q^{ll}} + g_p^{rl} g_q^{lr} \overline{g_p^{rr} g_q^{lr}} \right) + \\ & m^{rr} \left(g_p^{lr} g_q^{ll} \overline{g_p^{lr} g_q^{rl}} + g_p^{lr} g_q^{lr} \overline{g_p^{lr} g_q^{rr}} + g_p^{rr} g_q^{ll} \overline{g_p^{rr} g_q^{rl}} + g_p^{rr} g_q^{lr} \overline{g_p^{rr} g_q^{rr}} \right) + \\ & m^{lr} \left(g_p^{ll} g_q^{ll} \overline{g_p^{ll} g_q^{rl}} + g_p^{ll} g_q^{lr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rr} g_q^{rl}} + g_p^{rl} g_q^{lr} \overline{g_p^{rr} g_q^{rr}} \right). \end{aligned} \quad (3)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial m^{lr}} = & -d^{lr} g_q^{rr} \overline{g_p^{ll}} + \\ & -d^{ll} g_q^{rl} \overline{g_p^{ll}} + \\ & -d^{rr} g_q^{rr} \overline{g_p^{rl}} + \\ & -d^{rl} g_q^{rl} \overline{g_p^{rl}} + \\ & m^{rl} \left(g_p^{lr} g_q^{rl} \overline{g_p^{ll} g_q^{ll}} + g_p^{lr} g_q^{rr} \overline{g_p^{ll} g_q^{lr}} + g_p^{rr} g_q^{rl} \overline{g_p^{rl} g_q^{ll}} + g_p^{rr} g_q^{rr} \overline{g_p^{rl} g_q^{lr}} \right) + \\ & m^{ll} \left(g_p^{ll} g_q^{rl} \overline{g_p^{ll} g_q^{ll}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{lr}} + g_p^{rl} g_q^{rl} \overline{g_p^{rl} g_q^{ll}} + g_p^{rl} g_q^{rr} \overline{g_p^{rl} g_q^{lr}} \right) + \\ & m^{rr} \left(g_p^{lr} g_q^{rl} \overline{g_p^{ll} g_q^{rl}} + g_p^{lr} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rr} g_q^{rl} \overline{g_p^{rl} g_q^{rl}} + g_p^{rr} g_q^{rr} \overline{g_p^{rl} g_q^{rr}} \right) + \\ & m^{lr} \left(g_p^{ll} g_q^{rl} \overline{g_p^{ll} g_q^{rl}} + g_p^{ll} g_q^{rr} \overline{g_p^{ll} g_q^{rr}} + g_p^{rl} g_q^{rl} \overline{g_p^{rl} g_q^{rl}} + g_p^{rl} g_q^{rr} \overline{g_p^{rl} g_q^{rr}} \right). \end{aligned} \quad (4)$$

$$\begin{aligned}
\frac{\partial |S|^2}{\partial m^{\bar{l}}} = & -d^{lr} g_q^{lr} \overline{g_p^{\bar{l}}} + \\
& -d^{ll} g_q^{ll} \overline{g_p^{\bar{l}}} + \\
& -d^{rr} g_q^{lr} \overline{g_p^{rl}} + \\
& -d^{rl} g_q^{ll} \overline{g_p^{rl}} + \\
& m^{rl} \left(g_p^{lr} g_q^{ll} \overline{g_p^{\bar{l}}} \overline{g_q^{\bar{l}}} + g_p^{lr} g_q^{lr} \overline{g_p^{\bar{l}}} \overline{g_q^{lr}} + g_p^{rr} g_q^{ll} \overline{g_p^{rl}} \overline{g_q^{\bar{l}}} + g_p^{rr} g_q^{lr} \overline{g_p^{rl}} \overline{g_q^{lr}} \right) + \\
& m^{ll} \left(g_p^{ll} g_q^{ll} \overline{g_p^{\bar{l}}} \overline{g_q^{\bar{l}}} + g_p^{ll} g_q^{lr} \overline{g_p^{\bar{l}}} \overline{g_q^{lr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rl}} \overline{g_q^{\bar{l}}} + g_p^{rl} g_q^{lr} \overline{g_p^{rl}} \overline{g_q^{lr}} \right) + \\
& m^{rr} \left(g_p^{lr} g_q^{ll} \overline{g_p^{rl}} \overline{g_q^{rl}} + g_p^{lr} g_q^{lr} \overline{g_p^{\bar{l}}} \overline{g_q^{rr}} + g_p^{rr} g_q^{ll} \overline{g_p^{rl}} \overline{g_q^{rl}} + g_p^{rr} g_q^{lr} \overline{g_p^{rl}} \overline{g_q^{rr}} \right) + \\
& m^{lr} \left(g_p^{ll} g_q^{ll} \overline{g_p^{\bar{l}}} \overline{g_q^{rl}} + g_p^{ll} g_q^{lr} \overline{g_p^{\bar{l}}} \overline{g_q^{rr}} + g_p^{rl} g_q^{ll} \overline{g_p^{rl}} \overline{g_q^{rl}} + g_p^{rl} g_q^{lr} \overline{g_p^{rl}} \overline{g_q^{rr}} \right).
\end{aligned} \tag{5}$$